SYNOPSIS

ON

AUTOMATIC BUS SCHEDULING AND ROUTE MANAGEMENT SYSTEM

Submitted By:

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Submitted To:

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Title of the Project:

DCTSL : Automated Bus Scheduling And Route Management System

Objective:

The objective of this project is to develop a web-based Automated Bus Scheduling and Route Management System for Delhi City Bus. The system aims to automate bus scheduling, optimize route management, reduce delays, and improve resource utilization using modern web technologies.

Scope:

Scope Covered

1.Automated Bus Scheduling

- The system will automatically generate bus schedules to reduce manual work and errors.
- It will help in efficient allocation of buses and crew.

2.Route Management

- The system will allow management of bus routes.
- It will help in optimizing routes based on requirements.

3.GPS-Based Bus Tracking

- The system will track the location of buses using GPS technology.
- It will help transport managers monitor bus movement.

4.Resource Optimization

- The system will help in proper utilization of buses and crew.
- It will reduce idle time and improve service efficiency.

5.Improved Transport Management

- The system will improve reliability and service coverage.
- It will help transport authorities make better decisions.

6.Database Management

- The system will store information about buses, routes, schedules, and crew.

7.Web-Based Interface

- The system will provide a web interface for easy access and management.

8. Passenger Ticket Booking System

- The system will allow passengers to book tickets.

9.Online Payment Integration

- The system will include online payment features.

Scope Not Covered

1.Mobile Application

- The project will only be a web application, not a mobile app.

2.Direct Integration with Government Transport Systems

- The system will not connect to actual Delhi Transport Corporation servers.

3.Hardware Installation

- The project will not include installation of physical GPS devices.

4.Advanced AI Prediction

- The system will not use advanced artificial intelligence or machine learning models.

Methodology:

The project will follow the Software Development Life Cycle (SDLC).

Technologies Used:

Frontend:

- HTML
- CSS

- JavaScript

Backend:

- Node.js
- Express.js

Database:

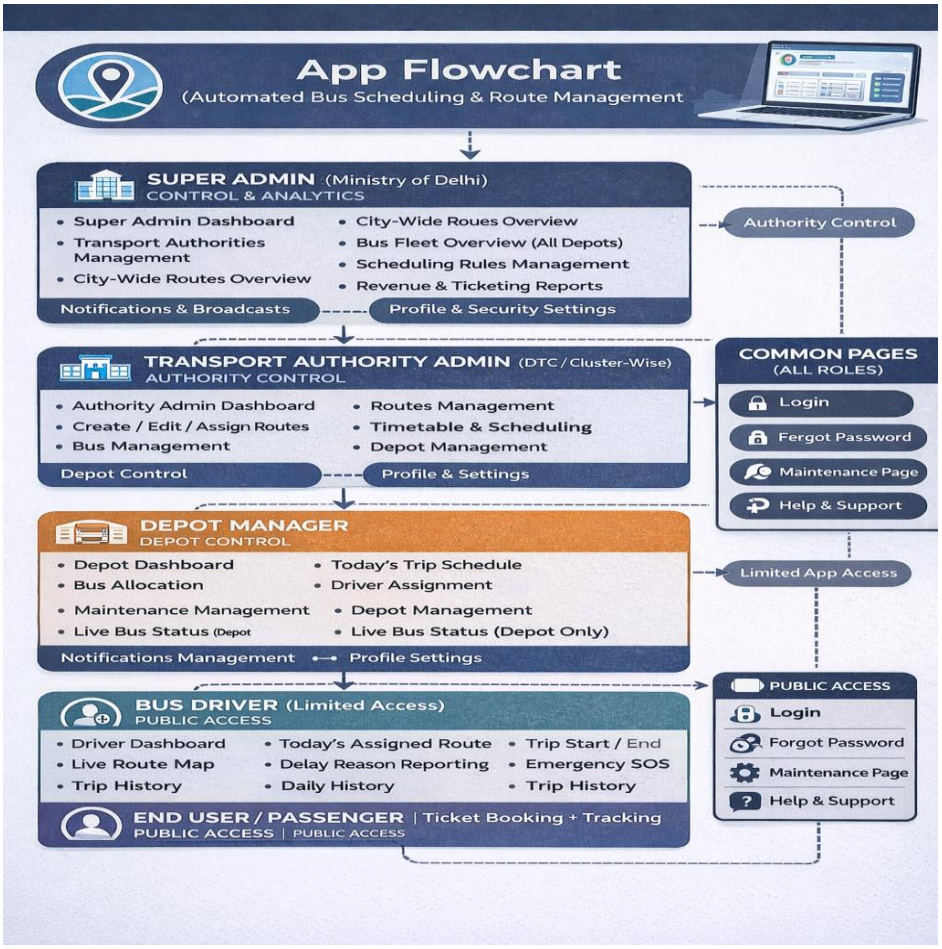
- MySQL / MongoDB

Map Integration:

- OpenStreetMap / Leaflet.js

Tools:

- VS Code
- Browser



Proposed System:

The Proposed System is a Web Application that automates bus scheduling and route management

Working:

- Admin logs into the system.
- Admin adds bus details and crew details.
- The system automatically assigns buses based on scheduling rules.
- Admin can create or modify routes using the map interface.

This System will reduce manual effort and Improve Efficiency

Features:

Automated Bus Scheduling

- The system automatically generates bus schedules to reduce manual work and improve efficiency.

GPS-Based Bus Tracking

- It tracks the real-time location of buses using GPS for better monitoring and control.

Real-Time Route Management

- The system manages and updates routes based on current traffic conditions.

Traffic-Based Route Optimization

- It uses real-time traffic data to reduce delays and improve route efficiency.

Route Visualization on Map

- It displays bus routes and locations on a digital map using OpenStreetMap/Leaflet.

Resource Utilization Management

- It helps in proper use of buses and crew to reduce idle time.

Database Management

- It stores and manages data related to buses, routes, schedules, and tracking.

Web-Based Dashboard

- It provides an interface for administrators to manage and monitor the transport system.

Implementation Plan:

Phase	Activity	Stack Involved	Timeline
1	Planning & Requirement Analysis	All	Week 1
2	System Design (Architecture, Database, Map Design)	MERN, Leaflet, OpenStreetMap	Week 2
3	Frontend Development	React.js	Week 3–4
4	Backend Development	Node.js, Express.js	Week 4–5
5	Database Setup	MongoDB, Mongoose	Week 4–5
6	GPS & Map Integration	Leaflet, OpenStreetMap	Week 5
7	System Integration	Full MERN Stack	Week 5
8	Testing & Debugging	MERN Stack	Week 6
9	Final Deployment & Documentation	MERN Stack, GitHub	Week 6
10	Maintenance	MERN Stack	Future Work

Team Members:

1. Harsh Agnihotri – Database Manager
2. Kesar Agrawal – Backend Developer
3. Akshat Agrawal – Backend Developer
4. Jatin Chaudhary – Frontend Developer
5. Devansh Chauhan – API & GPS Tracking Handler

Resources Required:

- Laptop / Computer
- Internet Connection
- VS Code
- Browser (Chrome / Edge)
- Node.js
- MongoDB
- React.js
- Express.js
- Map API (Leaflet / OpenStreetMap)
- Git / GitHub
- Postman

References:

- W3Schools – https://www.w3schools.com
- GeeksforGeeks – https://www.geeksforgeeks.org
- GitHub – https://www.github.com
- Leaflet Documentation – https://leafletjs.com
- Database System Concepts Book

Expected Outcomes:

- Fully working web application
- Automated bus scheduling system
- Route management with map integration
- Database integration using MongoDB
- User-friendly interface

Project Supervisor:

Under the guidance of: **Mr.Abhishek Singh**, Technical Trainer, Department of CEA.

Conclusion:

This project aims to develop an Automated Bus Scheduling and Route Management System for Delhi City Bus using the MERN stack. The system will automate bus scheduling, provide real-time route management, and integrate map-based tracking to improve efficiency and reliability. It will reduce manual work, minimize delays, and optimize resource utilization. The project will demonstrate the practical use of modern web technologies, databases, and map integration to solve real-world public transportation management problems and enhance the overall commuter experience.